

SNEHA V

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EDUCATION

Vellore Institute of Technology - BTech in Computer Science & Engineering

Aug 2023 - Expected 2027

TECHNICAL SKILLS

AI / ML: Python, PyTorch, TensorFlow, Scikit-learn, NLP, LLMs, Transformers, Speech Recognition, Deep Learning, Neural Networks, Prompt Engineering, Hugging Face, LangChain, RAG, Fine-tuning, Vector Databases

Development & Design: React.js, Next.js, TypeScript, Node.js, Tailwind CSS, HTML/CSS, Figma, UI/UX Design, REST API Integration

Tools: Git, GitHub, Jupyter Notebook, VS Code, Vercel, Prompt Engineering

EXPERIENCE

AI & ML Intern - Revoltution Labs Private Limited, Bangalore

May 2026 - Present

- Manual size-chart entry slowed garment onboarding, so I built an AI garment digitization pipeline using the Claude Vision API to auto-extract flat-lay measurements across 17 categories, achieving ± 1.5 -inch accuracy vs. physical measurements.
- Fragmented AI services risked vendor lock-in, so I designed a centralized Node.js AI Gateway with a provider-abstraction layer unifying virtual try-on (FASHN), multilingual translation (Claude), and a chatbot, enabling zero-downtime, vendor-agnostic model swapping.
- Manual digitization limited scalability for a luxury fashion e-commerce platform, so I engineered a ratio-based measurement system where a Vision LLM reads garment construction from one vendor image and computes precise measurement tables automatically.
- Hands-free interaction was missing from the product's voice workflows, so I built a voice assistant pipeline combining speech recognition, NLP, and LLM-based response generation, using prompt engineering to boost accuracy and response quality.

AI Automation Developer – KJ SYSTEM LTD

Apr 2026 - Jul 2026

- Manual receipt and invoice handling created bottlenecks in enterprise bookkeeping, so I built an Intelligent Document Processing solution using .NET 8 and Azure AI Document Intelligence, reducing manual effort by nearly 80% and streamlining financial data extraction.
- Inaccurate extracted data risked corrupting accounting records, so I developed a validation pipeline that verifies extracted financial and tax data before export, preventing bad records from entering accounting systems and improving reporting reliability.
- Disconnected OCR and reporting steps slowed document turnaround, so I implemented an end-to-end automation workflow integrating OCR, document intelligence, Excel report generation, and structured logging, enabling faster processing and scalable financial operations.

PROJECTS

Explainable Hybrid Clinical Decision Support System (CDSS)

[Project Link](#)

Python | Scikit-learn | MIMIC Datasets | XAI

- Built a hybrid CDSS combining Logistic Regression, Random Forest, and Decision Tree with rule-based clinical reasoning on MIMIC-III/IV and UCI datasets; evaluated on Accuracy, Precision, Recall, and F1-Score directly applicable to data analytics and AI reporting workflows.
- Developed an explainability layer providing feature-level insights and rule-based justifications, reducing false-positive alerts and improving interpretability of model outputs for practical decision-making.

Terminal Voice Agent - Hands-Free AI Coding Interface

[Project Link](#)

Python | Speech Recognition | LLM Integration | CLI

- Built a fully hands-free voice interface for AI terminal coding agents, enabling real-time speech-to-command execution - eliminating keyboard dependency for developer workflows.
- Integrated speech recognition with LLM-backed command interpretation; optimized latency and accuracy of the voice-to-action pipeline, gaining hands-on experience in end-to-end voice AI system design.

Breast Cancer Classification - Deep Learning (BreakHis Dataset)

[Project Link](#)

PyTorch | ViT | SHAP | Grad-CAM

- Benchmarked five deep learning architectures (ResNet50, VGG16, MobileNetV2, Custom CNN, Vision Transformer) on 7,909 histopathology images; ViT achieved 93.4% accuracy while MobileNetV2 provided the best efficiency/accuracy balance.
- Applied SHAP and Grad-CAM for model explainability, extracting clinical insights from AI predictions equivalent to AI-assisted data reporting and interpretability in business contexts.

YourFirstStep - Career Guidance Platform

[Project Link](#)

Next.js | TypeScript | Node.js | API Integration

- Built a full-stack AI-driven career guidance platform that helps students identify suitable career domains through interactive skill assessments and personalized recommendations, addressing the gap between academic learning and industry requirements.
- Developed a high-performance assessment engine with instant client-side evaluation and an interactive analytics dashboard, enabling real-time career insights while eliminating server-side processing delays.

CERTIFICATIONS

- Top 7 Finalist – Smart India Hackathon (SIH) 2025:** Selected among **540+ teams** for developing an AI-powered solution with a scalable backend and intuitive user experience.
- Hack With Hyderabad Finalist | Microsoft Campus:** Competed among **29,000+ participants** to build a **RAG-based AI Research Assistant** with real-time web retrieval, earning a finalist position and presenting at the **Microsoft** campus.